

Summer Class: Machine Learning and Data Science

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House Price Prediction

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The goal is to create a reliable system that can take these features for any house and produce an accurate price estimate.

House Price Prediction

Linear regression assumes the price is a weighted sum of its features.

Conceptually, the formula is:

$$\text{Price} = \text{Base Value} + (\text{Value per sq. ft.} \times \text{Size}) - (\text{Depreciation per year} \times \text{Age}) + (\text{Value per room} \times \text{Number of Rooms}) + \dots$$

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Price = Base Value + (Value per sq. ft. $\times$ Size) - (Depreciation per year $\times$ Age) + (Value per room $\times$ Number of Rooms) + ...

When dealing with thousands of houses, the most efficient way to represent this entire system is with the matrix equation:

$$y = X\beta + \epsilon$$

House Price Prediction

$$y = X\beta + \epsilon$$

- y : This is the price vector. It's a single column listing the known selling price for every house in your dataset. It's the “ground truth” the model learns from.

$$y = \begin{bmatrix} \text{Price}_1 \\ \text{Price}_2 \\ \vdots \\ \text{Price}_m \end{bmatrix}$$

House Price Prediction

$$y = X\beta + \epsilon$$

- X : This is the design matrix. Each row represents a single house, and each column represents a feature. A column of ones is included to account for the model's base value (the intercept).

$$X = \begin{bmatrix} 1 & \text{Size}_1 & \text{Age}_1 & \cdots & n\text{-th Feature}_1 \\ 1 & \text{Size}_2 & \text{Age}_2 & \cdots & n\text{-th Feature}_2 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & \text{Size}_m & \text{Age}_m & \cdots & n\text{-th Feature}_m \end{bmatrix}$$

House Price Prediction

$$y = X\beta + \epsilon$$

- β : This is the coefficient vector. This is what the model solves for. It contains the “rules” of the market—the specific monetary value associated with each feature (e.g., the value of an extra bedroom or the depreciation from another year of age).

$$\beta = \begin{bmatrix} \text{Base Value} \\ \text{Value per sq. ft.} \\ \text{Depreciation per year} \\ \vdots \end{bmatrix}$$

House Price Prediction

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- ϵ : This is the error vector. It's a list of the differences between the actual prices (y) and the prices predicted by the model ($X\beta$). No model is perfect, and this captures the unexplained part.

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- ϵ : This is the error vector. It's a list of the differences between the actual prices (y) and the prices predicted by the model ($X\beta$). No model is perfect, and this captures the unexplained part.

By solving this equation, regression finds the optimal coefficient vector (β) that makes the total prediction error as small as possible. You're then left with a powerful model that can price any house for which you have the features.

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- $X \in \mathbb{R}^{m \times n}$: design matrix.
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Optimization Problem:

$$\min_{\beta} \|y - X\beta\|_2^2$$

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Overdetermined vs Underdetermined Systems

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 - Infinite solutions.
 - This mathematical reality necessitates regularization to select a single, stable solution.

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- **Exact Solution:** No guessing or iteration is needed. It gives you the precise answer in one calculation.
- **No Hyperparameters:** You don't need to choose a learning rate or number of iterations.
- **Slow for Large Datasets:** Calculating the inverse of a correlation matrix is computationally very expensive.
- **Fails if Matrix is Not Invertible:** If features are multicollinear, the correlation matrix can't be inverted, and the equation fails.

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$$\beta_{k+1} = \beta_k - \alpha \nabla f(\beta)$$

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- Versatile: Can be used for many other machine learning algorithms, not just linear regression.

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- Versatile: Can be used for many other machine learning algorithms, not just linear regression.
- Requires Tuning: You must choose a learning rate (how big of a step to take). If it's too large, you'll overshoot the minimum; if it's too small, it will take forever to converge.
- Approximate Solution: It finds a value very close to the optimal β , but not the exact analytical solution.
- Can be Slow: May require many iterations to converge.

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- Provides Insights: The decomposition can reveal information about the structure of your data and the relationships between features.

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- Always Works: Unlike the Normal Equation, SVD works even if the correlation matrix is singular.
- Provides Insights: The decomposition can reveal information about the structure of your data and the relationships between features.
- Computationally Expensive: Like the Normal Equation, it can be slow for datasets with a huge number of features, though often more stable.

Overfitting vs Underfitting

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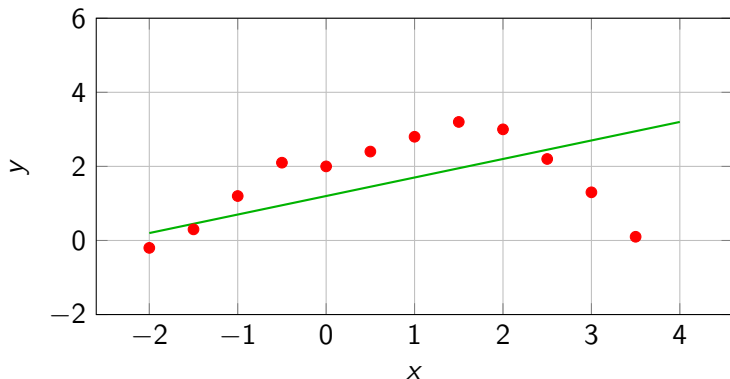
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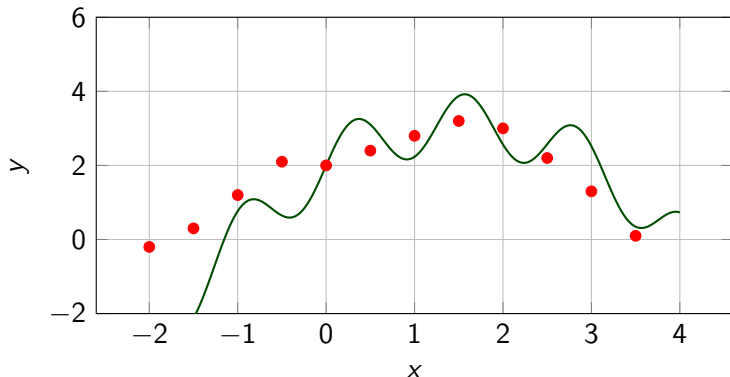
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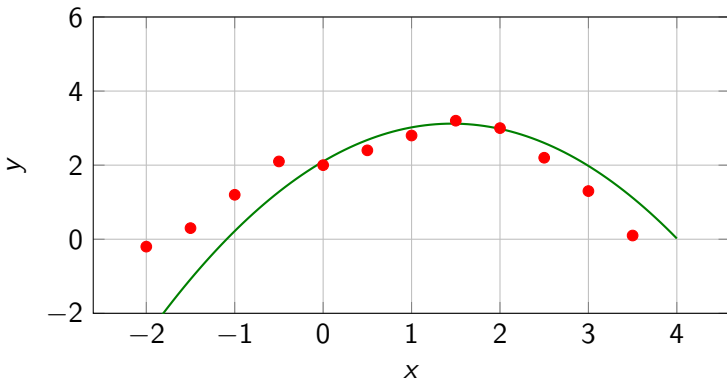


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- For a vector β with n elements, $\beta = [\beta_1, \beta_2, \dots, \beta_n]$, the p -norm is defined as:

$$\|\beta\|_p = \left(\sum_{i=1}^n |\beta_i|^p \right)^{1/p}$$

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- ① Take the absolute value of each element in the vector.
- ② Raise each absolute value to the power of p .
- ③ Sum all results.
- ④ Take p -th root of the sum.

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- l_1 -norm: Also known as the Manhattan norm or taxicab norm. It's the sum of the absolute values of the elements.

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- It encourages sparsity.
- By penalizing the sum of absolute values, the l_1 -norm finds it “cheaper” to push some coefficients all the way to zero rather than keeping them very small.
- Pushing a coefficient to zero is equivalent to removing that feature from the model entirely. This makes the l_1 -norm a powerful tool for automatic feature selection.

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Different values of p give us different, commonly used norms.

- l_2 -norm: This is the standard Euclidean norm, which corresponds to the intuitive, straight-line length of a vector.

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- It measures the straight-line distance from the origin.
- It shrinks all coefficients towards zero but does not force them to be exactly zero. This helps prevent overfitting by creating a simpler, less complex model that generalizes better to new data.
- It prefers smaller, more evenly distributed coefficients.

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- l_∞ -norm: Also known as the maximum norm or Chebyshev distance. As p approaches infinity, the norm simplifies to the maximum absolute value of any element in the vector.

$$\|\beta\|_\infty = \max(|\beta_1|, |\beta_2|, \dots, |\beta_n|)$$

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- The l_∞ -norm is used when the focus is on the single most extreme value rather than the overall magnitude.

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Different values of p give us different, commonly used norms.

- l_0 -norm: It is not technically a norm, but a function that counts the number of non-zero elements in a vector.

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- Finding the best subset of features is an NP-hard problem. It's computationally impossible to test every single combination of features for anything but the smallest datasets.
- The l_0 -norm is the theoretical ideal for creating a sparse model

Ridge Regression

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- A larger λ means a stronger penalty, forcing the model to use smaller coefficients.
- The primary effect of the l_2 penalty is shrinking the model's coefficients toward zero.
 - It reduces the influence of each individual predictor.
 - It does not set any coefficients exactly to zero. All features are kept in the final model.
 - This is especially useful when features are highly correlated (multicollinearity), as it helps to stabilize the model.

Ridge Regression

Cost Function:

$$J(\beta) = (y - X\beta)^T (y - X\beta) + \lambda \beta^T \beta$$

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To find the minimum, we compute the gradient of $J(\beta)$ with respect to β , denoted as $\frac{\partial J(\beta)}{\partial \beta}$.

The resulting gradient is:

$$\frac{\partial J(\beta)}{\partial \beta} = -2X^T y + 2(X^T X)\beta + 2\lambda \beta$$

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Set the gradient to zero to find the optimal coefficients, $\hat{\beta}$:

$$-2X^T y + 2(X^T X)\hat{\beta} + 2\lambda \hat{\beta} = 0 \Rightarrow \hat{\beta} = (X^T X + \lambda I)^{-1} X^T y$$

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- The l_1 penalty forces the coefficients of the least important features to shrink all the way to exactly zero.
 - If a feature's coefficient is zero, it's effectively removed from the model.
 - This creates a sparse model—a simpler model that only includes the most relevant predictors.
 - This makes LASSO a powerful tool for automatic feature selection.

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- We cannot simply take the derivative of the cost function and set it to zero to solve for β .
- The most common and intuitive algorithm for LASSO is **Coordinate Descent** and **Proximal Gradient Descent** (Lecture 1).

LASSO Regression

- For composite problems

$$\min_{\beta} F(\beta) = f(\beta) + g(\beta).$$

- $f(\beta)$: Convex, differentiable.
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- Proximal Operator:

$$\mathbf{prox}_{\alpha g}(v) = \operatorname{argmin}_{\beta} \left(g(\beta) + \frac{1}{2\alpha} \|\beta - v\|_2^2 \right).$$

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- For $g(\beta) = \lambda \|\beta\|_1$, **prox** is soft-thresholding:

$$\mathbf{prox}_{\alpha_k \lambda \|\cdot\|_1}(v_i) = \operatorname{sign}(v_i) \max(0, |v_i| - \lambda \alpha_k).$$

Comparison among Regular Regression, Ridge Regression and Lasso Regression

MATLAB Demonstration.

Summary of What You've Learned Today

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- Linear Regression with Regularization.
 - Vector Norm.
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See you on Monday (7/14) for Lecture 6 on
Sparse Regression and Dynamical Systems.